

Weekly Update

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The main update from this week was that Lucas and I participated in the ORCD workshop for distributed and parallel programming on MIT's Engaging system. Although most of the fundamentals were familiar to us, it was great to get hands-on experience working on parallelizing c programs on the thread level (using OpenMP) as well as for distributed systems with MPI, including compilation and execution through the terminal.

On the project front itself, the primary framework for single-node topology optimization has been completed (after Zhai et al.), as has that for ADMM. Although the ADMM system works great, the topology optimization still requires a few improvements as, while it works great to find topologies, it does not actually converge to a solution in the range of feasible stress (i.e. the material is overstressed relative to the requirements). This might be an issue of fine-tuning the parameters controlling this particular problem and it would be constructive to use some of the examples in Zhai et al. as a ground truth. This weekend will be a great time to work on that, and I will try to track down papers that list the adjoints for topology optimization with volume minimization (this particular example is still compliance minimizing instead of volume minimizing).